

ROSE BLANCHE GROUP



Specifications for the Acquisition of a MES for the subsidiaries of Rose Blanche Group

April 2024

1. Context:

In order to achieve optimal operational performance and enhance its market competitiveness, Rose Blanche Group is looking for partners to support it in the implementation of an MES system tailored to the specific needs of its subsidiaries. This project is a part of an initiative to adapt to market developments and fit in advances in technology and innovation.

2. Subsidiary Scope:

The objective of procuring a MES solution concerns multiple subsidiaries within the Rose Blanche Group, each operating in distinct sectors and presenting unique operational implications. Therefore, this initiative will exclusively focus on the industrial subsidiaries enumerated below:

- Cluster #1 (Agricultural):
 - ✓ ACN: Aliments Composés du Nord
 - A subsidiary specializing in the production of compound feeds for livestock
 - Head office: Nabeul
 - ✓ ALIFORT:
 - A subsidiary specializing in the production of Mash feeds.
 - Head office: Soliman
 - ✓ SOSEM: Société des Semences Sélectionnées
 - A subsidiary specializing in the production and multiplication of seeds and plants.
 - Head office: Tunis
 - ✓ STPA - ALCO: Société Tunisienne des Produits Alimentaires
 - A subsidiary specializing in the production of compound feeds for livestock.
 - Head office: Sfax
- Cluster #2 (Poultry):
 - ✓ Gallus:
 - A subsidiary specializing in the breeding of broiler chickens.
 - Head office: Mahdia
 - ✓ PISSO:
 - A subsidiary specializing in poultry breeding and the production of chicks and broilers.
 - Head office: Moknine
 - ✓ SAVINORD: Société Avicole du Nord
 - A subsidiary specializing in poultry breeding and the production of chicks and broilers.
 - Head office: Jendouba
 - ✓ SOPAT – MLIHA
 - A subsidiary specializing in the production and processing of chicken and turkey meat.
 - Head office: Sfax

- Cluster #3 (Cereals & transformation):
 - ✓ AMT: Agro Mélanges Technologies
 - A subsidiary specializing in the packaging of flour and semolina.
 - Head office: Sousse
 - ✓ CDS: Couscouseries du Sud
 - A subsidiary specializing in the production of couscous and pasta.
 - Head office: Sfax
 - ✓ GMCB: Les Grands Moulins du Cap Bon
 - A subsidiary specializing in the production of semolina and flour.
 - Head office: Nabeul
 - ✓ MCSR: Minoteries des Centre et Sahel Réunies
 - A subsidiary specializing in the production of semolina and flour.
 - Head office: Sousse
 - ✓ PATES WARDA
 - A subsidiary specializing in the production of pasta and couscous.
 - Head office: Sousse
 - ✓ SPPAS: Société De Production des Produits Alimentaires du Sud
 - A subsidiary specializing in the production of semolina and barley milling.
 - Head office: Sfax
 - ✓ STPA - Sboula: Société Tunisienne des Produits Alimentaires
 - A subsidiary specializing in the production of semolina and flour.
 - Head office: Sfax
 - ✓ FELEXOPRINT
 - A subsidiary specializing in the manufacturing and printing of packaging.
 - Head office: Sfax

3. Offer Composition:

The offer submitted by the provider must be comprehensive and structured into three main sections: an administrative file, a technical offer, and a financial offer. Each of these components includes key elements for a detailed evaluation:

- ✓ **Administrative File:** This file will assemble legal and administrative documents that confirm the compliance and reliability of the applicant's information, containing:
 - An extract from the commercial register, valid up to the offer reception deadline.
 - Reference attestations.
 - Examples of SLA contracts
- ✓ **Technical Offer:** Presenting a vision of the technical proposal with a detailed schedule and methodology:
 - For each of the requested service conditions, the applicant must specify how their technical offer meets the requested conditions.
 - Presentation of the roadmap (Service schedule)
 - Presentation of the project team with detailed resumes, references of similar projects.
- ✓ **Financial Offer:** The cost details associated with the service will be provided in accordance with the provisions of Appendix 3, including unit prices, VAT... signed with the company seal.

A technical offer and a financial offer must be presented for each pilot site by cluster, namely “CDS”, “SOPAT”, and “Pâtes Warda”. This means there will be three technical offers and three financial offers to be submitted for this contract.

The applicant will provide a comprehensive methodology outlining their approach to integrating the remaining subsidiaries of the Rose Blanche Group.

4. Functional Synthesis:

The table below will provide a comprehensive synthesis of the functionalities required for the solution, while specific details regarding functionalities by subsidiary and process will be presented in Appendix 6, offering a detailed and in-depth overview of functional requirements:

Specific functionality	Cluster		
	Agricultural	Poultry	Cereals & transformation
Identify the operator in charge of the workstation	✓	✓	✓
Gather and share real-time data the various parameters of the WO (WO status, progress rate, quantity produced on the WO, and material used)	✓	✓	✓
Report the quality test results of received materials from the LIMS ¹	✓		✓
Schedule the receipt of materials based on the status of storage locations (silos, buildings)	✓		✓
Schedule the preparation of ink mixtures			✓
Collect and communicate the status of each storage location (available, blocked, being cleaned, under maintenance)	✓		✓
Collect and communicate storage condition parameters (temperature, humidity)	✓		✓
Collect and communicate the conditions of the rearing environment (temperature, humidity, light intensity, CO2 levels, air pressure)		✓	
Automatically initiate adjustments to storage conditions if temperature or humidity exceed set limits	✓		✓
Collect and communicate on stored quantities (weight, allocation to work orders, quality data, location)	✓		✓
Collect and communicate the amount of waste or downgraded materials		✓	✓
Implement a deviation process to control the incorporation of raw materials between quality and production teams	✓		✓
Collect and communicate WO to various workstations from Sage X3	✓	✓	✓

¹ Laboratory Information Management System

Suggest a production schedule to the operator			✓
Record and communicate the time spent on each process	✓	✓	✓
Collect and communicate humidity levels during the process	✓	✓	✓
Track the quantity of products that are compliant, non-compliant, or under deviation	✓	✓	✓
Collect and communicate real-time updates on WO parameters (WO status, progress rate, quantity produced on the WO, and material used)	✓	✓	✓
Collect the quantity injected into the process (grains and derivatives, inks, water, chicks, eggs, and additives)	✓	✓	✓
Collect and share the recipe among formulation, production, and quality teams	✓		✓
Ensure a recipe validation workflow among relevant teams	✓		✓
Collect and communicate production quantities	✓	✓	✓
Provide real-time updates on quality test results	✓	✓	✓
Collect various metrics for performance indicator calculations	✓	✓	✓
Digitize all production and quality reports	✓	✓	✓
Collect and communicate machine counters to assist with preventive maintenance planning	✓	✓	✓
Automate the workflow for corrective maintenance from work request creation to equipment return post-repair	✓	✓	✓
Provide real-time updates on the status of different machines (operational or out of order)	✓	✓	✓
Share real-time machine parameters (speed, amperage, temperature, pressure)	✓	✓	✓
Alert and halt the manufacturing process if machine parameters exceed safe limits	✓	✓	✓
Automatically close the WOs	✓	✓	✓

Continuously collect and communicate parameters necessary for reporting	✓	✓	✓
Optimize the planning and scheduling of production by step and by sub-step	✓	✓	✓
Interact in Q&A mode for the consultation of operational documents	✓	✓	✓

5. Synthesis of Parameters to Collect

Data collection is of crucial importance in the implementation of a Manufacturing Execution System (MES), thus enabling process control and timely decision-making across various subsidiaries of Rose Blanche Group. Numerous parameters have been identified for collection throughout the various stages of manufacturing, as presented in the summary table below. It is important to note that this list is not exhaustive:

Parameters to monitor	Cluster		
	Agricultural	Poultry	Cereals & transformation
Quantity of cereals and derivative at the silo level (storage silo and buffer silo)	✓		✓
Temperature (building, silos, core, drying and cooling)	✓	✓	✓
Humidity (building, silos, wetting, drying and cooking)	✓	✓	✓
Quantity of cereals injected	✓		✓
Flow rate of injected cereal	✓		✓
Machine Amperage	✓	✓	✓
Number of bags produced	✓		✓
Amount of waste	✓		✓
Grinder rotation speed	✓		✓
Amount of additive injected (surfactant, palm oil, vitamins)	✓		✓
Voltage (Anesthesia & OAC Sorting Machine)		✓	

Frequency of anesthesia		✓	
Weight of the workpiece after evisceration		✓	
Weight of the workpiece in the grader		✓	
Number of Subjects Per Caliber		✓	
Batch number of penetrant carts		✓	
Calibration of Penetrant Testing Carts		✓	
Entry and exit time (testing chamber, fumigation chamber, incubator, drying chamber, cooking machine)		✓	✓
Elapsed time during the process	✓	✓	✓
Quantity rejected due to foreign object detection	✓	✓	✓
Downgraded quantity in the cutting chamber		✓	
Product density in the clipper		✓	
Pressure in the Cooking Machine		✓	✓
Number of chick mortalities		✓	
Number of chicks		✓	
Chick Weight		✓	
Light Intensity		✓	
CO2 levels		✓	
Depression		✓	
Number of eggs produced		✓	

Number of Broken Eggs		✓	
Number of chicks hatched		✓	
Volume of water injected into the process		✓	✓
Quantity of feed injected into the process		✓	
Water consumption rate		✓	✓
Machine cycle rate	✓		✓
Quantity of product under deviation	✓		✓
Percentage of opening of the unloading hoppers			✓
Compliant Product Quantity	✓		✓

6. Technical Abstract for MES Implementation

This technical abstract outlines the key components and requirements for the implementation of an MES (Manufacturing Execution System). It encompasses the current application architecture, target technological framework, communication protocols and interfacing mechanisms, as well as the proposed structure of the MES solution.

This MES implementation aims to streamline production operations by harmonizing data exchange and providing comprehensive monitoring and control capabilities, boosting overall production efficiency across the manufacturing environment.

7. Current Application Architecture:

The existing architecture includes an Enterprise Resource Planning (ERP) system, along with Computer-Aided Production, Maintenance, Quality, and Energy Management Systems. These systems collectively manage various aspects of production operations.

8. Target Technological Framework:

The target framework involves seamless interfacing between the ERP and the MES, as well as integration with Siemens, Omron, and Mitsubishi programmable logic controllers (PLCs). Furthermore, interfacing with Quality Management Systems, Maintenance Systems, Energy Management Systems, and Production Systems is essential for holistic operational management.

9. Communication Protocols and Interfacing:

Communication protocols and interfacing mechanisms play a critical role in facilitating data exchange and system integration. The proposed protocols include OPC UA, Node-RED, MQTT, API integration, Ethernet/IP, and serial communication protocols such as RS-232 and RS-485. These protocols enable interoperability between disparate systems and devices.

10. Structure of the Target MES Solution:

The MES solution must be designed with a robust architecture to support the integration of information flows across the production environment. This includes the establishment of interfaces with ERP and various operational systems. Additionally, the MES solution encompasses features such as real-time data acquisition, production scheduling, quality management, maintenance tracking, and energy consumption monitoring.